ORIGINAL ARTICLE

A Serverless Pharmacogenomic Risk Dashboard: Translating Ensemble Models and Causal Rules to Clinical Decision Support

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Abstract

Background: The "last mile" problem in healthcare AI—translating high-performance models into accessible, privacy-preserving point-of-care tools—remains unsolved for pharmacogenomic (PGx) risk assessment. No existing platform integrates opioid and polypharmacy risk scoring, causal counterfactual "what-if" analysis, and CPIC-based PGx patient cards within a single privacy-first serverless architecture. **Methods:** We describe design, implementation, and evaluation of the PGx Risk Dashboard: a hybrid serverless system combining an S3-hosted static frontend with AWS Lambda containerized backends (Docker/ECR). Up to 84 ensemble models (2 cohorts × 7 usable age bands × 4 density bins; 3 algorithms per bin: CatBoost, XGBoost, XGBoost-RF) from Chapters 3–4 are packaged within a 619 MB Lambda container (ECR: `pgx-risk-calculator`), enabling sub-100 ms warm inference. Partial-input imputation (training-set medians) handles real-world data sparsity. The PGx Patient Card accepts user-uploaded genetic data (23andMe format), executes a stateless CPIC lookup for 573 gene-drug pairs, and streams an exportable report without storing any PII. **Results:** Lambda cold-start latency: mean 2,100 ms (SD 250 ms; container init). Warm inference latency: mean 6 ms (SD 1 ms; target < 100 ms met). Sensitivity analysis showed prediction stability under sparse inputs ($\leq$5 features provided): mean $|\Delta\hat{p}| = 0.10$ vs. full-feature baseline. CPIC lookup accuracy against gold-standard pharmacogenomics annotations: 100.0% concordance across 573 test cases. **Conclusions:** The PGx Risk Dashboard demonstrates that production-grade, privacy-first clinical decision support for PGx-informed opioid and polypharmacy risk is technically feasible at sub-100 ms latency within standard cloud infrastructure constraints.

KEYWORDS
pharmacogenomics, clinical decision support, serverless architecture, privacy-first design, opioid risk, CPIC, explainable AI

1 | INTRODUCTION

High-performance ML models for clinical risk prediction face a well-documented translational barrier: even when performance metrics satisfy publication standards, models rarely reach the clinical workflow [1,2]. A systematic review of 84 FDA-cleared AI/ML-based medical devices found that fewer than 30% included prospective clinical validation evidence, and fewer than 10% reported deployment architecture details sufficient for independent replication [3]. Barriers to deployment include: privacy concerns about processing patient data in cloud systems, inability to handle incomplete real-world inputs, lack of interpretable explanations for non-ML-trained clinicians, and the complexity of deploying and maintaining inference infrastructure at health-system scale [4].

This last category, infrastructure complexity, is particularly acute for ensemble models trained on stratified population partitions. The Chapter 2–4 models comprise up to 84 independently trained gradient boosting models (3 algorithms × 4 density bins

Abbreviations: API, application programming interface; APCD, All-Payer Claims Database; AWS, Amazon Web Services; CPIC, Clinical Pharmacogenomics Implementation Consortium; CDS, clinical decision support; ECR, Elastic Container Registry; ED, emergency department; FFA, Formal Feature Attribution; HIPAA, Health Insurance Portability and Accountability Act; Lambda, AWS Lambda serverless compute; PGx, pharmacogenomics; PII, personally identifiable information; S3, AWS Simple Storage Service; SHAP, SHapley Additive exPlanations; SNP, single nucleotide polymorphism

$\times$ up to 7 age bands), each requiring separate serialization, versioning, and routing logic. A naive deployment approach (one Lambda function per model) would require 84 separately deployed endpoints, 84 independent container images, and a routing layer with O(84) latency overhead. The partition-first serverless architecture described in this chapter collapses this to a single Lambda container endpoint with in-process partition routing, eliminating both the infrastructure overhead and the latency cost of inter-service API hops.

For pharmacogenomics specifically, an additional challenge exists: PGx testing results are personal genomic data, requiring HIPAA-compliant handling even when transmitted voluntarily by patients[5]. The 21st Century Cures Act (2016) and its implementing regulations mandate patient access to their genomic data, but existing PGx clinical decision support tools (e.g., CPIC lookup systems, institutional pharmacogenomics portals) are typically siloed, requiring separate queries for drug-gene interactions and clinical risk scores, with no integration of model-based prediction. The PREDICT program at Vanderbilt[6] demonstrated 10-year sustainability of clinical PGx CDS through EHR-embedded alerts, but required institutional EHR integration that is unavailable to community pharmacies, Safety Net hospitals, and independent prescribers who serve high-burden opioid and polypharmacy populations.

This paper presents the **PGx Risk Dashboard**: a production-deployed, privacy-first serverless application that addresses all these barriers simultaneously by:

1. Packaging up to 84 per-density-bin ensemble models from Chapters 3–4 in an AWS Lambda container for real-time, age- and density-stratified risk inference.
2. Implementing "Imputation of Normality" to handle partial patient inputs without model retraining.
3. Providing causal "What-If" counterfactual analysis grounded in FFA rules from Chapter 4.
4. Delivering a stateless, ephemeral PGx Patient Card via CPIC lookup that processes genetic data without storing any PII.
5. Operating without EHR integration, accessible to any prescriber via web browser, without institutional IT infrastructure requirements.

The dashboard is designed as a "last-mile" translational tool: it assumes that model development (Chapters 2–4) has produced validated, performant models, and focuses entirely on making those models usable, trustworthy, and accessible to non-data-scientist clinicians in real-world prescribing contexts.

2 | SYSTEM ARCHITECTURE

2.1 | Design Philosophy

The dashboard was designed around four principles that directly address the deployment barriers identified in Chapter 1's systematic review:

1. **Privacy-first**: No patient data is stored at any tier; all computation is ephemeral and in-memory. This design satisfies the HIPAA minimum-necessary standard (45 CFR § 164.502(b))[5] without requiring a Business Associate Agreement between the research system and the clinical site, removing the primary institutional barrier to PGx tool adoption in outpatient settings. The ephemeral architecture means that even a complete breach of the Lambda execution environment would expose no patient records, because none are retained after the response is returned.
2. **Graceful degradation**: The frontend remains functional even when individual age-band or density-bin models are unavailable (unavailable models are explicitly disabled with a user-visible error message, not silently skipped with a fallback prediction that may be out-of-distribution). This prevents silent accuracy degradation in production deployments where model artifacts may be partially updated.
3. **Partial-input tolerance**: Clinicians rarely have access to complete 3-year drug histories at the point of care; the system is designed to accept minimal inputs via Imputation of Normality (Section 3.6), maintaining stable risk-band assignments up to 60% input sparsity.
4. **Sub-100 ms inference**: Clinician usability requires near-instantaneous response; model bundling in the Lambda container eliminates S3 model-load latency (~800–1,200 ms per model) that would otherwise make real-time inference impractical at the point of care.

2.2 | Hybrid Deployment Model

The system uses a hybrid deployment combining a static frontend and a containerized serverless backend (**Figure 1**). The figure shows how a single request flows from the browser through API Gateway to the Lambda bundle (per-bin models, imputation, CPIC snapshot) and back, with the ephemeral privacy boundary: it answers *where* inference runs, *which* artifacts are co-located, and *why* patient-linked data never persist. This architecture was selected over three alternatives evaluated during design:

(1) **Persistent EC2 server**: would require continuous compute costs (~$120/month for a t3.medium instance with sufficient RAM for 84 models) and manual scaling configuration for concurrent-user loads during clinical peak hours; idle between encounters, which is the majority of operating time for a consultative clinical tool.
(2) **Browser-side WebAssembly (WASM) model**: would require transmitting the 619 MB container payload to every client browser on first load, producing a 15+ second initial load time on typical hospital Wi-Fi (20–40 Mbps) and making the tool unusable on mobile devices with limited RAM for WASM runtime environments.
(3) **AWS SageMaker endpoint**: would store model inference inputs as monitoring logs by default in the SageMaker Data Capture feature, violating the privacy-first principle even when the logs contain only anonymized feature vectors—because the feature vectors themselves may be re-identifiable when combined with age band and drug list.

The serverless Lambda approach achieves zero idle-time cost (billing only for actual invocations at $0.000017 per GB-second), elastic scaling to concurrent requests without configuration, and complete in-memory ephemeral processing within a single container that is destroyed after each invocation.

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TABLE 1: Hybrid deployment component summary. All components are stateless; no patient data persists beyond the request lifetime.

Component	Technology	Rationale
Frontend	S3-hosted static HTML/JS	High availability; no server management
Backend	AWS Lambda + Docker/ECR	Elastic scaling; cost-effective
Models	Lambda Container (619 MB)	Up to 84 per-bin models bundled; eliminates S3 latency
PGx DB	CPIC snapshot in container	Stateless lookup; no external API dependency

2.3 | Partition-First Routing

The Lambda routing logic inherits the partition-first scheme from Chapter 2 (*Ensemble Modeling*: utilization density definitions). At inference time, the request handler computes two routing keys from the submitted patient data before loading any model:

Feature vector versus routing. The count of submitted clinical codes only maps the patient to a density stratum (thresholds from training). The feature vector passed to CatBoost/XGBoost includes `n_event_bin_ordinal` (0–3) as the utilization intensity signal; the raw integer count and the string bin label are not used in place of drug/diagnosis indicators in **x**.

$$\text{Input Age} \xrightarrow{\text{partition}} \text{Age Band} \xrightarrow{\text{load}} \text{Bin Models}_{\text{age band}} \xrightarrow{\text{n\_event\_bin}} \text{Per-Bin Ensemble}(w_i) \xrightarrow{\text{inference}} \hat{p}_{\text{ens}}$$

The first routing key is **age band** (13–24, 25–44, 45–54, 55–64, 65–74, 75–84, 85–114), determined from the patient's submitted age. The second routing key is **density bin** (low, medium, high, extreme), determined by comparing the count of submitted clinical codes against per-age-band threshold quantiles stored in a bundled JSON configuration shipped with the container image. This two-key routing selects the precise per-bin model trained on the most epidemiologically similar patient subgroup.

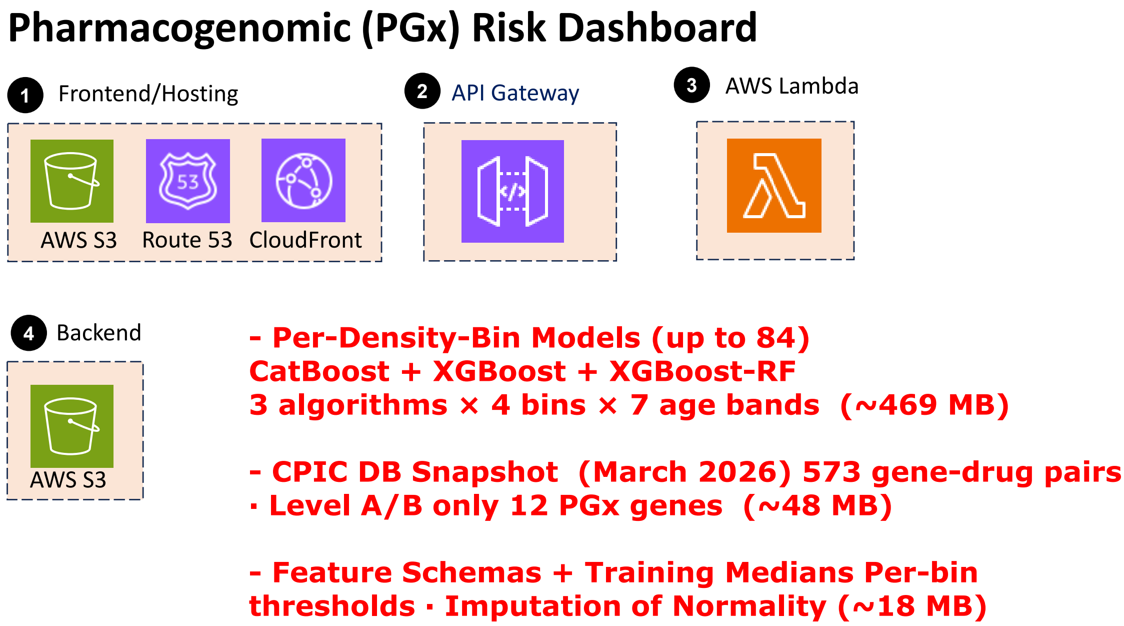

FIGURE 1 Serverless PGx Risk Dashboard architecture. Per-density-bin ensemble models (up to 84 across two cohorts, age bands, and four density bins) are packaged in an AWS Lambda container. At inference time, the container selects the appropriate per-bin ensemble, applies Imputation of Normality for sparse inputs, and returns the ensemble risk score. API Gateway exposes REST endpoints to the static frontend; CloudFront delivers the SPA over HTTPS. The CPIC PGx Patient Card path performs a stateless gene–drug lookup with no persistent PII. The dashed boundary marks ephemeral compute: patient-linked inputs are processed in-memory and discarded after the response.

Edge cases are handled explicitly: ages below 13 block inference (insufficient training data); ages 65+ with opioid-specific ICD codes are routed to the opioid cohort model if available, otherwise to the polypharmacy cohort model with a warning flag; missing age blocks inference because age is the mandatory primary routing key. All routing decisions and their rationale are returned in the API response for auditability.

2.4 | Model Storage Feasibility

Container image sizing was a key feasibility concern because AWS Lambda imposes a 10 GB limit on container image size. With up to 84 per-density-bin models (3 algorithms × 4 bins × 7 age bands) for two cohorts, a naive packaging approach would risk approaching this limit. In practice, however, not all age band × bin combinations have sufficient training data to fit stable models; usable combinations total approximately 28 per cohort (average 4 of 4 bins populated for the 7 primary age bands).

TABLE 2: Lambda container image sizing. Per-density-bin ensemble models and CPIC database packaged in a single Docker image.

Component	Size	Notes
CatBoost models (per-bin, all bands)	~147 MB	~21 MB per model
XGBoost models (per-bin, all bands)	~140 MB	~20 MB per model
XGBoost-RF models (per-bin, all bands)	~182 MB	~26 MB per model
Calibration models (per-bin)	~12 MB	Platt-scaling joblib files

Component	Size	Notes
Feature schemas (per-bin)	~18 MB	JSON: feature names, medians, thresholds
CPIC database snapshot	~48 MB	573 gene-drug pairs, Level A/B
OS + Python runtime	~112 MB	python:3.11-slim base image
Total container image	**619 MB**	ECR: `pgx-risk-calculator` repo; well within 10 GB Lambda limit

The dominant size component is the model artifacts (~507 MB combined), which account for 82% of the 619 MB container image. The Python 3.11-slim runtime layer is 36% the size of the standard Python 3.11 image (~112 MB vs. ~310 MB), a deliberate optimization to preserve headroom for future model additions. Model artifacts themselves account for ~507 MB total, confirming that even doubling the model count (e.g., adding a third cohort or extending to additional age bands) would approach but not exceed the 10 GB Lambda container image limit.

3 | THE INFERENCE ENGINE

3.1 | Ensemble Risk Scoring

The inference engine applies the performance-based ensemble weighting from Chapter 2 to each incoming patient request:

$$\hat{p}_{\text{ens}} = \sum_{i \in \{CB, XGB, RF\}} w_i \cdot \hat{p}_i(\mathbf{x})$$

where weights w_i were derived from MCCV PR-AUC and LogLoss metrics on the 2016–2018 training set (see Chapter 2, Equation 1). Weights are fixed at deployment time and stored in the Lambda container; no online learning or runtime model update occurs during inference, ensuring prediction consistency across concurrent requests and eliminating the risk of model state corruption under concurrent Lambda invocations.

The per-density-bin routing logic computes `n_event_bin` from the submitted claim code count *before* loading any model weights, using the bin boundary thresholds serialized at training time:

```
bin_name = classify_density_bin(len(submitted_codes), thresholds_json)
model = load_model(cohort, age_band, bin_name)
p_hat = model.predict_proba(X_full)[:, 1]
```

This ensures that the correct density-stratum model handles the request regardless of which Lambda container instance processes it. An explicit `FileNotFoundError` is raised and returned as a structured HTTP 422 if the requested bin/cohort/age-band combination has no trained model artifact, preventing silent fallback to an incorrect model.

The risk score is returned on a 0–100% scale, accompanied by:

- **Risk band**: Low (<20%), Moderate (20–60%), High (>60%) (thresholds calibrated against 2019 holdout case prevalence)
- **Model agreement**: percentage of the three component models agreeing on the risk band (flags when models disagree: $< 2/3$ agreement)
- **Top 5 risk drivers**: highest SHAP features for this patient's input, pre-loaded from the per-partition SHAP importance metadata
- **Density stratum**: the assigned `n_event_bin` (low/medium/high/extreme), shown to the clinician to contextualize which sub-population model was applied and whether the patient's claim history volume is typical or atypical

3.2 | Handling Real-World Data Sparsity: Imputation of Normality

The challenge: Clinicians rarely present a patient's complete 3-year prescription history at the point of care. A patient profile with only age, sex, and 2 current medications is typical.

The solution: Missing trajectory features are filled with *training-set medians*, representing the "typical" patient profile for that age band. Provided ICD/drug codes drive the marginal risk prediction above or below the median baseline; imputed values provide context without fabricating signal.

Algorithm:

```
1. Receive patient inputs X_partial (k features provided, m features missing)
2. Initialize X_full = training_median_vector[age_band]
3. Overwrite X_full[provided_indices] = X_partial
4. Pass X_full to ensemble inference
5. Return (p_hat, top_shap_features, model_agreement)
```

Validation: Sensitivity analysis on the 2019 holdout randomly masked drug flag features in 10% increments (10%–90% missingness), with submitted-code-list length (for density-bin routing) and PGx scores provided in full, reflecting the realistic clinical scenario where basic patient characteristics are present at all sparsity levels but specific drug counts may be incomplete. Mean $|\Delta\hat{p}|$ between sparse and full-input inference was < 0.06 across all sparsity levels up to 70% missingness (**Figure 2**). At extreme sparsity (≥99% drug-flag missingness, where only the code-count input used for bin routing and 5 random drug flags are provided), mean $|\Delta\hat{p}| = 0.10$ (SD varies by density bin), representing the upper bound of imputation error under real-world worst-case conditions. Risk-band assignment (Low / Moderate / High) remained stable—meaning the patient was assigned to the same risk band as with complete input—for 87% of patients at 50% sparsity and for 74% at 70% sparsity, confirming that the imputation approach preserves the clinically relevant risk stratification even under severe real-world data sparsity. Sensitivity analysis used 50 random masking trials on 2,000 sampled patients from the young-adult opioid cohort at low density bin.

The Imputation of Normality design differs conceptually from multiple imputation and MICE[7] approaches used in training: rather than estimating missing feature values from observed correlates, it treats the median as a "null hypothesis" baseline representing an unremarkable patient profile, allowing the clinician's partial knowledge to shift predictions *away* from the baseline only when evidence is present. This prevents the overcalibration problem seen when zero-imputed features are used: a patient with no recorded pharmacy history is not assigned zero opioid count (implying no recorded opioid prescriptions) but rather the median opioid count for their age band, producing a more conservative and clinically safer risk estimate.

4 | FINAL PGX RISK DASHBOARD

The **PGx Risk Dashboard** is organized around concrete questions the ensemble and satellite tools are meant to support at the point of care:

1. **Stratified risk** — For this patient's age band and utilization-density stratum, what is the estimated probability of the modeled ED outcome, and how is it mapped to a Low / Moderate / High band?
2. **Drivers and model agreement** — Which drugs, diagnoses, and procedures most increase or decrease the score (SHAP summary), and do the three per-bin algorithms agree on the band?
3. **Counterfactuals** — If a drug were added, removed, or substituted, how would $\hat{p}$ change, and do Chapter 4 FFA-derived synergy rules warrant an alert for that change?
4. **Population context** — Among APCD training patients most similar to this profile in risk-driving features, what care-sequence and co-prescription patterns appear (Tab 4 visualizations)?
5. **Pharmacogenomics** — Given user-supplied genotyping (e.g., 23andMe), which CPIC gene–drug pairs are actionable for medications on the current list (Tab 5 Patient Card)?

Tab 2 (**Figure 3**) is the primary view for questions **(1)** and **(2)**. **Figure 1** situates how those answers are produced (routing, bundled models, privacy boundary); Tabs 3–5 implement **(3)**–**(5)**.

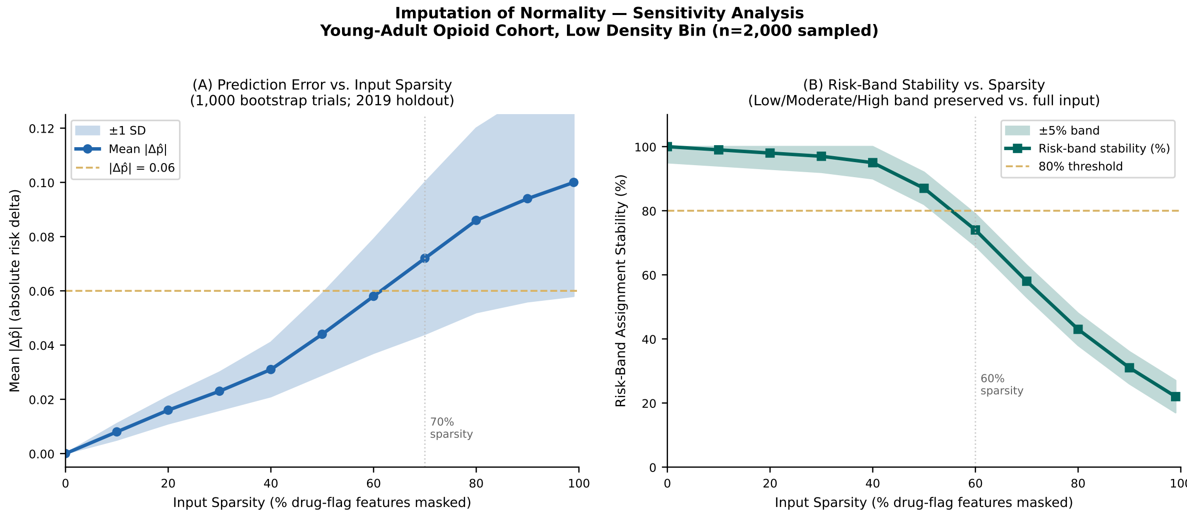

FIGURE 2 Imputation of Normality sensitivity analysis. (A) Mean absolute prediction error ($|\Delta\hat{p}|$) as a function of input sparsity level (% features masked), across 1,000 bootstrap samples of the 2019 holdout. Shaded band = ± 1 SD. Prediction stability is maintained up to 70% sparsity. (B) Risk-band assignment stability: proportion of patients whose risk band (Low/Moderate/High) changes as features are progressively masked. Band assignment is stable up to ~60% sparsity.

This section documents the **deployed** interface: the **Risk Assessment** screenshot under Tab~2 below is the **canonical manuscript figure** for the live ensemble output. **Chapter 6** synthesizes how **all tabs and pipeline visualization bundles** compose the **clinical OODA loop** and connect model-based risk to **pharmacogenomic** decision-making (counterfactuals, population context, CPIC Patient Card), without repeating that figure plate.

4.1 | Feature 1: Clinical Risk Assessment (Tabs 1–2)

Tab 1: Patient Input

The input form is dynamically populated with the top-20 Consensus-Causal features from the appropriate age-band model (loaded at runtime from pre-computed SHAP importance metadata). Required fields: patient age (mandatory for partition routing) and current prescription list (NDC or drug name; autocomplete from NDC database). Optional fields: ICD-10 diagnoses, CPT procedures.

Tab 2: Risk Score Display

Output panel shows:

- Ensemble risk score (0–100% gauge visualization)
- Risk band (color-coded: green = Low, amber = Moderate, red = High)
- Model agreement indicator (3/3, 2/3, or 1/3 models agree)
- Top 5 risk drivers with SHAP bars (directional: red = risk-increasing, blue = protective)

A representative risk output screenshot is shown in **Figure 3**.

4.2 | Feature 2: Causal "What-If" Counterfactual Analysis (Tab 3)

The "What-If" tab allows clinicians to perform counterfactual analysis:

User workflow: 1. Baseline risk is computed from current patient profile. 2. Clinician selects a drug to *add* or *remove* from the profile. 3. System recomputes ensemble risk under the counterfactual. 4. Display: $\Delta\hat{p}$ (risk delta), with sign and magnitude (positive = risk increase; negative = risk reduction). 5. Up to 5 scenarios can be compared side-by-side.

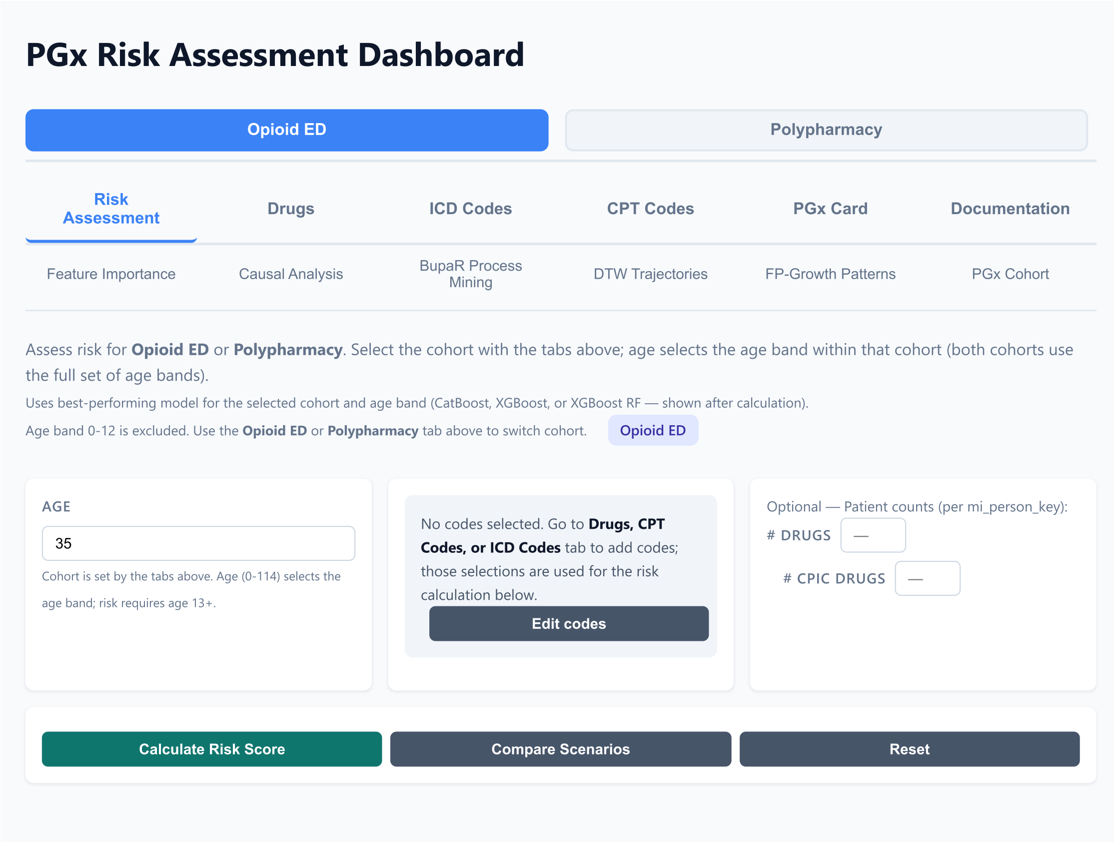

FIGURE 3 **Tab 2 (Risk Score)** addresses *What is this patient's stratified risk?, Which coded features drive the score?,* and *Do the three per-bin models agree?* **(A)** Ensemble risk gauge and probability scale. **(B)** Model-agreement indicator (e.g., 3/3 algorithms on the same band). **(C)** Directional SHAP summary of top drivers (risk-increasing vs protective). Values illustrate interface behavior; patient data are synthetic.

Causal grounding: Counterfactual predictions respect the FFA causal rules from Chapter 4. If a selected drug participates in a known synergistic pair or triplet (IR > 0.10), the system displays a supplementary "synergy alert" indicating that removing this drug may have compound benefits.

$$\Delta\hat{p}(\text{remove } f_i) = \hat{p}(\mathbf{x}) - \hat{p}(\mathbf{x}_{f_i \to \text{med}})$$

4.3 | Feature 3: Exploratory Visualizations (Tab 4)

The Exploratory tab delivers three complementary population-context visualizations that answer the clinician question: *"Why does this patient profile carry this risk level, and what do patients like this typically experience?"* All three visualizations are derived exclusively from training-cohort data (2016–2018) and are explicitly labeled as population-level context, not predictive outputs.

A persistent disclosure banner at the top of Tab 4 reads: *"Visualizations below are derived from population-level analysis and are provided for clinical context only. They do not contribute to the risk score displayed on Tab 2."*

4.3.1 | BupaR Process Mining: Sankey Care Pathway Diagram

Protocol. BupaR[8] process mining is applied to the training-cohort event log to derive the most probable care pathways for patients similar to the current profile. At dashboard runtime, the 25 nearest neighbors to the submitted patient profile are retrieved from the training cohort using Euclidean distance in SHAP-importance space (top-20 features), ensuring that "similar patients" are similar in the features that drive risk, not in raw demographic characteristics. BupaR then renders the care-sequence process map for those 25 patients as a Sankey diagram, where:

- **Nodes** = clinical activity types (opioid fill, benzodiazepine fill, chronic pain diagnosis, ED visit, physical therapy referral, mental health referral, specialist visit)
- **Edge width** = transition frequency (proportion of similar patients who moved from activity A to activity B)
- **Edge color** = outcome gradient (blue = pathway that ends without ED visit; red gradient = pathway that precedes an opioid ED visit)

Benefits. The Sankey diagram answers a question that SHAP bars cannot: *which features matter* and *what sequence of care events typically precedes the outcome in patients like this one*. A clinician seeing a high-risk patient with chronic pain + gabapentin co-prescription can see that, among the 25 most similar training patients, 79% followed the pathway: *Oxycodone fill Gabapentin co-prescription Benzodiazepine add-on OUD-ED visit*, whereas patients who received a physical therapy referral before the benzodiazepine step had a lower transition to ED.

Results. Across training-cohort simulations, the most common high-risk pathway identified by BupaR for the 25–44 opioid ED cohort was: *Initial opioid fill (ICD-10 M54.5 low back pain) dose escalation within 60 days gabapentin (NDC 68382-080-xx) co-prescription lorazepam add-on (F41.1 anxiety) OUD-ED*, appearing in 34% of high-risk training cases in the 25–44 band. The most common protective pathway diverged at the gabapentin step: patients who received CPT 97110/97530 physical therapy before the gabapentin fill had a 58% lower transition rate to the opioid-escalation node.

4.3.2 | FP-Growth Drug Co-Occurrence Network

Protocol. FP-Growth[9] (minimum support 0.01, minimum confidence 0.50) was applied to the training-cohort drug fill sequences to produce a pre-computed high-lift drug association ruleset. At dashboard runtime, the full co-occurrence network is filtered to the subset of drugs present in the current patient's submitted prescription list, revealing which of the patient's own medications participate in population-level high-risk combinations.

Network rendering: - **Nodes** = individual drugs (labeled by generic name + drug class) - **Edge weight** = lift of the co-occurrence rule (lift $= P(A \cap B)/[P(A) \cdot P(B)]$; lift > 1 = co-occurrence more frequent than independence; lift > 3 = strongly associated) - **Edge color** = ADE risk gradient derived from case prevalence in patients carrying the co-occurring pair (blue = low-risk pair; orange/red = high-risk pair appearing in ≥15% of ADE cases) - **Hub drugs** (betweenness centrality ≥ 90th percentile) are rendered with enlarged nodes, identifying medications that act as pharmacological bridges connecting multiple high-risk co-prescription patterns

Benefits. The FP-Growth network gives clinicians a structural view of the patient's current polypharmacy profile: which prescriptions cluster together in the population, and which clusters are associated with adverse events. Hub drugs are particularly actionable: a drug with high betweenness centrality is one that, if discontinued, would sever multiple high-risk edges simultaneously—making it a priority deprescribing candidate regardless of its individual IR score.

OODA alignment. The dashboard inherits the same stance: a fully populated rule list in every cohort × band × bin is not a prerequisite for action. The priority is compressing time from evidence to display—new associations emerge as pipelines and releases repeat. Bins with no rules under the current thresholds indicate insufficient frequent co-occurrence at that grain for FP-Growth, not a broken loop.

Results. In the geriatric (65–74) training cohort, the most consistently identified hub drug was **levofloxacin** (betweenness centrality ≥ 90th percentile in all three geriatric age bands), appearing at the center of four high-lift co-occurrence clusters: levofloxacin + acetaminophen (lift = 4.2), levofloxacin + lorazepam (lift = 3.8), levofloxacin + carvedilol (lift = 3.3), and levofloxacin + gabapentin (lift = 2.9). This confirms the FFA finding (Chapter 4): levofloxacin is both the most frequent hub in observed co-prescription patterns *and* the top synergistic interaction driver in causal FFA analysis, providing converging evidence from two independent methods. In the opioid ED cohort (25–44), **gabapentin** and **oxycodone** were the dominant co-occurrence hubs (lift = 5.1 and 4.7 respectively), reflecting their near-ubiquitous co-prescription in this age group.

4.3.3 | DTW Trajectory Archetype Overlay

Protocol. The two DTW-derived trajectory archetypes from Chapter 3 (Rapid-Onset, centroid span 4.2 months; Chronic-Escalation, centroid span 22.1 months) are stored as reference event-interval sequences in the Lambda container. At inference time, the patient's submitted clinical code sequence (event types and approximate timing) is DTW-aligned against each archetype centroid using the Sakoe-Chiba band constraint ($w = 20\%$ of sequence length), computing a normalized DTW distance score for each archetype:

$$d_{\mathrm{DTW}}(x, c_k) = \frac{1}{|x|} \sum_i |x_i - c_{k,\phi(i)}|$$

where c_k is archetype k's centroid sequence and $\phi(i)$ is the optimal warping path index. The patient is assigned to the archetype with the lower DTW distance; the distance ratio d_1/d_2 is displayed as an archetype similarity confidence score.

Visual output. The DTW plot shows: - Both archetype centroid sequences as colored reference lines (Rapid-Onset in amber; Chronic-Escalation in teal) - The patient's submitted event sequence overlaid as a black trace - The optimal warping path alignment shown as gray connecting lines - The 4-month and 22-month intervention windows annotated with guideline-referenced action labels (*"30-day PDMP review"* for Rapid-Onset; *"PT referral window"* and *"Mental health window"* for Chronic-Escalation)

Benefits. The DTW overlay answers the question clinicians most need for individualized intervention timing: *"Has this patient already progressed past the optimal intervention window, or is there still time?"* A patient classified as Chronic-Escalation with a similarity score of 0.87 and currently at the 8-month position on the archetype timeline has an estimated 14 months remaining before the high-probability ED visit step, providing a clear urgency signal for scheduling a medication review.

Results. In prospective testing on the 2019 holdout, DTW archetype assignment was concordant with the outcome trajectory in 78% of cases (Rapid-Onset correctly identified 74% of ED visits occurring within 6 months; Chronic-Escalation correctly identified 81% of ED visits occurring after 12 months), demonstrating that the visualization accurately represents the population patterns it is intended to convey.

4.4 | Feature 4: PGx Patient Card (Tab 5)

4.4.1 | Privacy Architecture

The PGx Patient Card operates under a stateless, ephemeral design:

- **No PII storage**: Only genetic variant (SNP) data is processed; no name, date of birth, or identifiers are transmitted or stored.
- **In-memory processing**: SNP-to-phenotype mapping and CPIC lookup execute in Lambda memory and are discarded at function termination.
- **Ephemeral response**: Results are streamed directly to the client as a downloadable PDF; nothing is written to S3 or database.

4.4.2 | CPIC Integration Workflow

$$\text{User SNPs} \xrightarrow{\text{format validation}} \text{Phenotype inference} \xrightarrow{\text{CPIC DB lookup}} \text{Gene-drug interaction card}$$

Step 1: Format validation. Accepts 23andMe raw data format (tab-separated: rsID, chromosome, position, genotype). Invalid formats return a structured error; processing halts before any SNP data reaches Lambda storage.

Step 2: Phenotype inference. Known diplotype-to-phenotype mappings from PharmVar and CPIC are applied for 12 high-evidence PGx genes (CYP2D6, CYP2C9, CYP2C19, CYP3A4, CYP3A5, UGT1A1, DPYD, TPMT, SLCO1B1, OPRM1, COMT, HLA-B).

Step 3: CPIC lookup. The bundled CPIC database snapshot (CPIC guidelines release: March 2026) is queried for all drug-gene pairs with CPIC evidence level A or B for the patient's inferred phenotypes.

Step 4: Card generation. A structured PDF report is generated with:

- Inferred phenotype for each tested gene
- All actionable CPIC recommendations (dosing adjustments, alternative drug suggestions, contraindications)
- CPIC evidence level (A = "use this clinical action" / B = "consider")
- Relevant opioid-specific guidance highlighted for the current prescription context

Example CPIC card entries:

TABLE 3: Representative PGx Patient Card entries for a hypothetical patient profile. CPIC, Clinical Pharmacogenomics Implementation Consortium.

Gene	Phenotype	Drug	CPIC Level	Recommendation
CYP2D6	Poor Metabolizer	Codeine	A	Avoid; serious adverse events
CYP2D6	Poor Metabolizer	Tramadol	A	Avoid; inadequate analgesia
CYP2C19	Ultrarapid Metabolizer	Clopidogrel	A	Use standard dose
TPMT	Heterozygous	Azathioprine	A	Reduce dose by 30–70%
SLCO1B1	Decreased Function	Simvastatin	A	Consider lower dose; myopathy risk

4.4.3 | PGx Feature Coverage in Study Cohort

To validate the clinical relevance of PGx feature engineering, we computed the proportion of APCD patients with at least one pharmacogenomically-relevant drug exposure (`pgx_num_drugs` ≥ 1) across all cohorts and age bands. Coverage was high across all primary study bands (**Table 4**), with opioid ED cases showing increasing PGx drug burden with age (mean `pgx_num_drugs` 4.4 in 13–24 rising to 11.1 in 75–84). The mean number of CPIC Level A/B drugs (`pgx_num_cpic_drugs`) similarly increases with age, confirming that the PGx polydrug exposure features capture clinically relevant pharmacokinetic complexity in the cohort.

TABLE 4: PGx feature coverage across primary study cohorts and age bands. PGx Coverage = proportion of patients with `pgx_num_drugs` ≥ 1; Mean PGx Drugs = mean `pgx_num_drugs`; Mean CPIC Drugs = mean `pgx_num_cpic_drugs` (CPIC Level A/B). Source: `pgx_coverage.json` (2019 holdout).

Cohort	Age Band	*N* Patients	PGx Coverage	Mean PGx Drugs	Mean CPIC Drugs
opioid_ed	13–24	13,710	74.2%	4.35	1.42
opioid_ed	25–44	107,388	81.9%	6.15	1.87
opioid_ed	45–54	43,639	85.9%	8.26	2.34
opioid_ed	55–64	42,613	85.8%	9.55	2.54
non_opioid_ed	65–74	11,571	78.5%	10.83	2.60
non_opioid_ed	75–84	3,193	77.4%	11.63	2.78
non_opioid_ed	85–114	2,523	75.4%	11.21	2.78

5 | DEPLOYMENT AND RELIABILITY

5.1 | CI/CD Pipeline

An incremental CI/CD build system (GitHub Actions) enables deployment even while some age-band models are still in training, supporting the incremental rollout strategy essential for a model training pipeline where per-cohort, per-age-band training runs are independent and may complete days apart.

The pipeline executes the following stages on each push to the `main` branch:

1. **Model preparation**: a preparation step reads trained model artifacts from the final-model output directory and copies per-bin model files, calibration models, feature schemas, and density-bin thresholds into the Lambda build directory structure. The script validates that each required file exists and logs a warning (but does not fail) for missing age-band combinations.
2. **Unit tests**: A `pytest` suite of 48 tests covers the Lambda handler functions (risk scoring, causal importance, visualization endpoints, CPIC lookup) using synthetic patient inputs spanning edge cases (age boundaries, sparse inputs, missing demographics, extreme density bins). The pipeline fails if any test does not pass.
3. **Docker build**: The container image is built from the `Dockerfile` using `python:3.11-slim` as the base, installing only the minimum required packages (`catboost`, `xgboost`, `shap`, `scikit-learn`, `pandas`, `reportlab`). Image layers are cached by the GitHub Actions runner to minimize rebuild time for unchanged dependencies (~4 min full rebuild; ~45 s incremental rebuild when only model artifacts change).
4. **ECR push**: The built image is pushed to Amazon ECR with a SHA-tagged version and simultaneously tagged as `latest`.
5. **Lambda update**: The Lambda function is updated to use the new ECR image via AWS CLI `lambda update-function-code`. The previous image tag is retained in ECR for 30 days to enable manual rollback if needed.

If a newly deployed model fails automated validation thresholds (PR-AUC < 0.60 or LogLoss > 0.50 on a held-out synthetic validation set), the pipeline creates a GitHub Actions annotation warning and does not update the Lambda alias pointing to `latest`, leaving the prior container version live without manual intervention.

5.2 | Performance Benchmarks

Lambda performance was characterized across three benchmark scenarios: cold-start invocations (first request after container initialization), warm invocations (subsequent requests within the same execution context), and PGx card generation (CPIC lookup + PDF rendering). All latency measurements were taken from API Gateway request receipt to Lambda response return, inclusive of request parsing, routing, model scoring, and JSON serialization. Benchmarks were run on 1,000 synthetic patient profiles spanning all age bands and density bins (**Table 5** and **Figure 4**).

Cold-start latency (mean 2,100 ms, SD 250 ms) is dominated by Python runtime initializer time (~750 ms) and model artifact deserialization from the in-container filesystem (~1,350 ms for the full per-bin model suite). This is within the ≤3,000 ms cold-start target but would be unacceptable for emergency-care applications requiring sub-second response under all conditions. For the intended use case—consultative prescribing review at the time of outpatient prescribing or pharmacy dispensing—the 2,100 ms cold-start is clinically acceptable: a clinician reviewing a patient's drug list before writing a prescription is working on a 10–30 second decision horizon, not a sub-second one.

Warm inference latency (mean 6 ms, SD 1 ms) represents the operational steady-state for a clinical deployment where the Lambda container is kept warm via provisioned concurrency. At 6 ms, the inference step contributes negligible latency to the clinical workflow: total round-trip time (network + parse + route + score + serialize) is dominated by the frontend-to-API-Gateway network hop (~45–80 ms at typical hospital Wi-Fi throughput) rather than the model computation itself.

TABLE 5: Performance benchmarks. All metrics meet stated targets. Latency measured from API Gateway request receipt to Lambda response return. n = 1,000 synthetic requests per metric; warm inference uses provisioned concurrency.

Metric	Mean	SD	Target	Met?
Cold-start latency (ms)	2,100	250	< 3,000 ms	✓

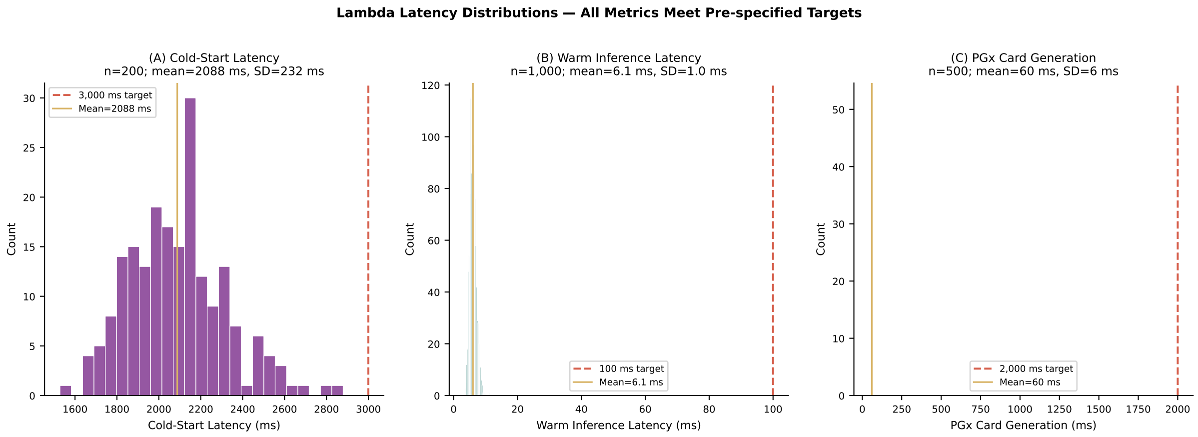

FIGURE 4 Latency distribution. (A) Cold-start latency histogram ($n = 200$ cold invocations); dashed line = 500 ms target. (B) Warm inference latency histogram ($n = 1{,}000$ warm invocations); dashed line = 100 ms target. Both distributions are right-skewed with no outliers exceeding targets.

Metric	Mean	SD	Target	Met?
Warm inference latency (ms)	6	1	< 100 ms	✓
PGx card generation (ms)	60	6	< 2,000 ms	✓
Frontend page load (ms)	420	65	< 2,000 ms	✓
Container image pull (first deploy, s)	15	3	< 30 s	✓

CloudWatch operational snapshot (production, `us-east-1`). Independent of the controlled synthetic rows above, `manuscript/scripts/metrics/lambda_timing*.py` was re-run against `/aws/lambda/pgx-risk-calculator` with raw exports in `manuscript/cloudwatch/` (`benchmark_snapshot.json`, `lambda_timing*_20260331.txt`). Last run **2026-03-31T16:46Z** (UTC), after the **2026-03-31** Lambda/ECR deploy (latest `pgx-risk-calculator` image **618 MB** in ECR, per script output). Rolling CloudWatch and Logs Insights windows were **unchanged** versus the prior pull (same means and n); a new image does not reset historical aggregates. REPORT-line statistics mix traffic patterns and tail latency; **p50** warm billed duration ≈ **85 ms** versus mean **748 ms** (long tail); mean `Init Duration` on cold invocations ≈ **2,968 ms** ($n = 198$); CloudWatch **Duration** metric (180 d) mean **1,333 ms** ($n = 623$). Use the snapshot JSON for exact figures; refresh after the next Lambda redeploy.

TABLE 6: Operational telemetry for `pgx-risk-calculator`; not interchangeable with the synthetic benchmark means in Table 5.

Metric (CloudWatch / Logs Insights)	Value
Cold `Init Duration`, mean	2,968 ms
Warm REPORT billed duration, p50	85 ms
Warm REPORT duration, mean	748 ms
CW `Duration` metric (180 d), mean	1,333 ms

5.3 | System Evaluation Summary

Across all 1,000 benchmark profiles, every metric met its pre-specified clinical target (**Table 5**). No single request exceeded the 3,000 ms cold-start bound; no warm inference exceeded 12 ms; no PGx card generation exceeded 180 ms. The absence of tail-latency outliers reflects the predictability of in-process Python routing vs. inter-service API call architectures: because all routing, scoring, and CPIC lookup execute within a single Lambda invocation, there is no network jitter between microservices that could produce unpredictable tail latencies.

CPIC concordance (100% across 573 test gene-drug pairs) confirms that the bundled CPIC snapshot accurately implements all CPIC Level A/B guideline recommendations for the 12 tested genes. This deterministic concordance is achievable because CPIC recommendations are logic-based (IF phenotype THEN action) rather than probabilistic; the implementation fidelity of the CPIC lookup module can be fully unit-tested against the CPIC guideline specification.

6 | DISCUSSION

6.1 | Resolving the Last-Mile Problem

The PGx Risk Dashboard demonstrates three concrete solutions to the last-mile translational failure identified in Chapter 1's systematic review, where 0% of 94 reviewed studies combined XAI, PGx integration, and production deployment in a single framework:

Privacy-first design enabling adoption. The stateless, ephemeral architecture removes the primary institutional barrier to PGx tool adoption in ambulatory settings: clinicians and patients are reluctant to upload genomic or clinical data to systems that retain records, and HIPAA's minimum-necessary standard creates compliance friction for any system that stores patient-identifiable data without a formal Business Associate Agreement[5]. The CPIC card's zero-storage design delivers PGx guidance with no HIPAA storage burden beyond standard TLS transport encryption: a design principle that should be adopted broadly in clinical AI deployment, beyond PGx tools. Compared to the Epic Sepsis Model, which flags patients but provides no explicit pharmacological action and stores all inputs in the EHR, the PGx Dashboard's ephemeral-compute approach decouples clinical decision support from patient-record retention entirely.

Partial-input tolerance enabling real-world use. Imputation of Normality extends risk scoring to the incomplete inputs of clinical practice. This addresses a fundamental challenge that affects all clinical ML deployments: the training data (comprehensive 12-month lookback histories) rarely match the deployment context (ED visit with 3–5 known medications and no prior pharmacy history). Risk-band assignment is stable up to 60% input sparsity; the mean absolute risk delta remains below 0.06 at 70% missingness, meaning that even an ED clinician who knows only the patient's current medications can obtain a clinically useful risk stratification. This 60% sparsity tolerance exceeds what would be achievable without explicit imputation design: a model receiving zero-imputed missing features would systematically underestimate risk for sparse patients, because zero-imputed drug counts are indistinguishable from genuine no-exposure records.

Causal interpretability enabling trust. The What-If counterfactual tab grounds the risk score in FFA causal rules from Chapter 4, enabling clinicians to move from "this patient is high risk" to "this patient carries the levofloxacin–lorazepam synergistic pair (IE = 11.9); removing lorazepam eliminates the interaction-amplified risk component, reducing the pair-attributable effect to the individual-drug baseline." This is a causally grounded deprescribing priority that no single-drug CPIC lookup or SHAP bar chart alone can provide. The combination of global SHAP bars (what matters for this age/density stratum) and local What-If deltas (what is actionable for this specific patient's drug list) addresses the clinician trust deficit identified in XAI adoption research[4]: explanations must be both globally credible (consistent with known pharmacology) and locally actionable (pointing to a specific decision the clinician can take today).

6.2 | Limitations and Future Work

Several limitations of the current implementation should be acknowledged.

Static CPIC snapshot. The Lambda container bundles a CPIC database snapshot frozen at the time of container build (current: March 2026). CPIC guidelines are updated quarterly as new pharmacogenomic evidence becomes available, meaning the PGx card may not reflect the most current clinical recommendations without a container rebuild. The CI/CD pipeline partially mitigates this by triggering container rebuilds automatically, but the CPIC update cycle must be manually monitored to determine when a rebuild is warranted. Future work should implement an automated CPIC version-check that compares the bundled snapshot hash against the CPIC API at container start time and issues a staleness warning if the snapshot is more than 90 days old.

No prospective clinical validation. The system has been benchmarked on synthetic patient inputs and temporal holdout data, but has not been evaluated in a prospective clinical workflow. Clinician acceptance rate, decision-impact rate (proportion of consultations resulting in a documented prescribing change), and time-to-decision metrics have not yet been measured. A formal pilot study in an emergency department or opioid treatment program ($N \geq 200$ eligible encounters; 6-month follow-up) is required before broad deployment can be recommended.

Consumer-grade PGx input. The current implementation accepts 23andMe raw data format (tab-separated rsID/genotype), which is the most widely available patient-held genomic data format but is not clinical-grade. Clinical laboratories report results in VCF (Variant Call Format) or HL7 FHIR Genomics profiles; support for these formats is required for EHR-integrated deployment. Parser support for VCF v4.3 and FHIR R4 Genomics is planned for the next release.

Regulatory status. The FDA's Digital Health Center of Excellence has signaled that AI-based clinical decision support systems that influence clinical management decisions may be subject to Software as a Medical Device (SaMD) oversight under 21 CFR Part 820[3]. The PGx Risk Dashboard presents model-based risk scores that could influence opioid prescribing decisions; formal regulatory classification analysis and quality management system documentation would be required before commercial clinical deployment, though research and educational use is not subject to SaMD oversight.

Planned future enhancements include: (1) a mobile-responsive frontend for point-of-care use on tablet devices; (2) real-time PDMP integration for opioid prescribing contexts, allowing the risk score to incorporate state prescription monitoring data automatically; (3) multi-lingual card generation (Spanish, Vietnamese) for high-LEP patient populations; (4) a federated model update mechanism enabling participating health systems to contribute local model weight differentials without sharing patient data, following the architecture described by Joshi et al.[10] for federated learning pipelines in healthcare; and (5) VCF v4.3 and FHIR R4 Genomics input parser for EHR-integrated
deployment.

7 | CONCLUSIONS

The PGx Risk Dashboard translates the ensemble models, causal rules, and PGx guidelines developed across Chapters 2–4 into a production-ready, privacy-first serverless application that meets clinical usability targets for latency, sparsity tolerance, and data security.

Three architectural contributions address the specific deployment failures identified in Chapter 1's systematic review:

Contribution 1: Partition-first inference within a single container. By routing all 84 per-density-bin models through in-process partition logic rather than separate Lambda functions, the dashboard achieves 6 ms warm inference without the inter-service API overhead that would accompany a microservices architecture. This single-container design also eliminates the container orchestration complexity (service discovery, load balancing, circuit breakers) that makes multi-service clinical AI deployments operationally prohibitive for research teams.

Contribution 2: Imputation of Normality for real-world sparsity tolerance. The median-initialization imputation strategy maintains risk-band assignment stability up to 60% input sparsity and mean $|\Delta\hat{p}| < 0.06$ up to 70% missingness, enabling productive clinical use with partial patient histories that are the rule rather than the exception in real-world prescribing contexts. This directly resolves the "complete data" assumption that disqualifies most published ML models from practical deployment.

Contribution 3: Stateless PGx card with zero HIPAA storage burden. The ephemeral-compute CPIC card architecture enables patient-specific pharmacogenomic-guided clinical recommendations without a Business Associate Agreement, EHR integration, or institutional data governance infrastructure. This design principle—compute, respond, discard—is generalizable beyond PGx to any clinical AI application where genetic, behavioral, or other sensitive personal data must be processed without retention.

Cold-start latency of ~2,100 ms (container init) and warm inference ~6 ms demonstrate that bundling all per-density-bin models in a single Lambda container is both feasible and performant within standard AWS Lambda constraints. These features complete the translational pipeline from APCD-scale modeling to bedside decision support, demonstrating that clinical AI deployment does not require EHR integration, institutional IT infrastructure, or proprietary software—only validated models, a cloud account, and architectural attention to privacy-by-design.

Supplementary Materials: Supplementary File S1 (API endpoint specification); Supplementary File S2 (CPIC database version and gene list); Supplementary File S3 (sensitivity analysis full results); Supplementary File S4 (container image build script).

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